Kim 2020 Lung Adenocarcinoma — analysis summary

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

13/13 malignant-bearing samples intermix on one continuous manifold with no per-patient islands (radius-graph: 0 detached; iLISI median 1.00→2.75 of 13); both compartments (Embryonic 9,315 / Cycling 1,197) carry the trajectory.

Gate 2 — module geography → GO (non-branching)

Single Gast→Crest developmental axis: Gast declines along pseudotime (ρ=−0.55) from the Gast/pSom-high origin; Crest anti-correlated with Gast (r=−0.55), rising to a terminal readout (ρ=+0.48; Crest-high decile mean pt=0.81). pSom overlaps the Gast body (ρ=+0.09, flat) — an inverse gradient, not a Gast-arm/pSom-arm co-expression hub. Crest terminal (not a Gast arm) — no inversion flag.

Gate 3 — hub mutual exclusivity → NA

Single monotone axis — no bifurcation and no Mixed-type hub, so there is no branch pair to test for mutual exclusivity.

STEP FIGURES

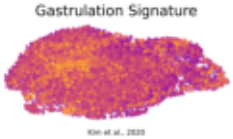
Gate 1: integration



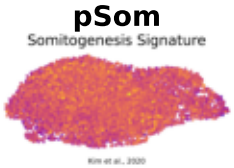
Trajectory / root



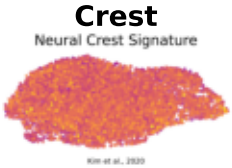
Gast



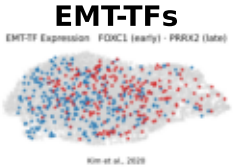
pSom



Crest



EMT-TFs



Dataset. LUAD (Kim et al., 2020), 10x — 14 patients / 32,493 cells (full atlas); malignant only: 10,538 cells → 10,512 across 13 samples enter preprocessing (drop 1 sub-42 spillover sample, P0008/26 cells); the whole malignant manifold (no lobe scoping) enters the trajectory.

Verdict. ✓ Non-branching pattern reproduces — Gate 1 GO, Gate 2 GO, Gate 3 NA. Kim's LUAD malignant manifold is a single monotone gastrulation→terminal-neural-crest gradient rooted at the Gast/pSom-high origin, matching the developmental axis; it lacks Puram's hub-and-spoke bifurcation, so Gate 3 has no hub to test. The Embryonic/Cycling split is partly proliferative, but the developmental modules are cell-cycle-orthogonal (|ρ vs g2m|<0.15; both compartments share identical module means), so the whole manifold is kept (Bischoff/Sun no-scoping precedent), carrying pseudotime↔cycling co-variation (ρ=+0.41) as a declared caveat. EMT-TFs, detection-gated (>3%) over a Snail/Zeb/Twist + FOXC/PRRX panel: FOXC1 (early/root) → PRRX2 (late/terminus) — the only two factors that clear the detection floor, chosen by data over Puram's pair.

Parameter choices

Min cells/sample	42 (gap 70→26, r=2.69)	**Integration**	Harmony θ=5; PCA50; spread=1.3
Scope	whole manifold (no scoping)	**Trajectory**	scFates curve N=20 λ=0.3 μ=100
Root node	tip 0 = Gast/pSom-high, Crest-low	**Compartments**	Embryonic 9,315 / Cycling 1,197